

Four Strategies, 562 Trades, Zero Edge

A Forensic Autopsy of Algorithmic Weather Betting

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Abstract

We document the systematic failure of four algorithmic trading strategies on Polymarket weather prediction markets, reporting 509 live trades over six weeks (February–March 2026) that collectively lost \$401 on a \$372 starting balance. Each strategy generation attempted to solve the predecessor’s failure mode: ensemble forecasting (v4, 38 trades, 22% WR), smart-money copy-trading (v6, 206 trades, 63% WR), elite wallet replication (v7, 35 trades, 25% WR), and multi-wallet convergence signals (v8, 230 trades, 36% WR). None achieved positive returns. Meanwhile, 482 paper trades under five backtested configurations produced \$172 in profits at 71% win rate—a gap we trace to an information-theoretic limit: forecast mean absolute error (2.03°F) equals the contract bucket width (2°F), rendering probabilistic edge irreducible regardless of ensemble size or strategy sophistication. We identify a *profit-lock paradox* specific to binary markets: the configuration with the highest win rate (79.7%) earned the most profit, but only because it held positions to resolution—every strategy using intermediate profit-taking or stop-losses destroyed value, because exiting a \$0.20 binary position at \$0.30 captures just 12.5% of maximum gain while retaining 100% of maximum loss. On-chain analysis of 37,562 wallets reveals that exploitable edge does exist: 276 elite wallets maintained 97.8% win rates, likely through latency arbitrage on the resolution source (Weather Underground) rather than superior forecasting. The 144% growth in market participation over 10 days eroded smart wallet win rates from 84.4% to 81.1%, consistent with rapid professionalization. These results establish that Polymarket weather markets are weak-form efficient for systematic retail strategies despite persistent elite exploitation.

Keywords: prediction markets, weather forecasting, algorithmic trading, smart money, market efficiency, Polymarket, binary options, backtest-to-live gap

1 Introduction

Prediction markets have long been studied as mechanisms for information aggregation [Gelman et al., 2020, Lockhart, 2025]. Weather prediction markets on Polymarket represent a particularly clean test of market efficiency: outcomes are externally verifiable, the underlying information is publicly available, contracts resolve within 24–48 hours, and blockchain infrastructure makes all trades publicly observable.

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These characteristics create an appealing opportunity for systematic trading. Public ensemble weather forecasts from ECMWF (51 members) and GFS (31 members) provide probabilistic predictions. On-chain transparency enables identification of historically profitable wallets. The short resolution cycle allows rapid strategy iteration. Over six weeks, we developed and deployed four distinct algorithmic strategies, each designed to address the failure mode of its predecessor:

1. **v4:** 82-model ensemble forecasting with Student- t probability mapping and Kelly sizing
2. **v6:** Smart-money copy-trading with diversified NO positions across multiple buckets
3. **v7:** Elite wallet replication targeting only the highest-performing on-chain traders
4. **v8:** Multi-wallet convergence signals requiring 3+ independent wallets to agree

All four lost money. This paper presents the forensic analysis of why, drawing on 509 live trades, 482 paper trades, 34 backtested configurations, and a wallet ecosystem dataset covering 37,562 participants. Our results are relevant beyond weather markets: the profit-lock paradox, the information-theoretic forecasting limit, and the two-tier efficiency structure we document apply to binary prediction markets generally.

Contributions

- We present the first multi-strategy empirical study of systematic weather prediction market trading, reporting 509 live trades across four strategy generations with complete trade-level data.
- We identify an information-theoretic limit: when forecast MAE equals contract bucket width, no ensemble approach can generate edge, regardless of model count.
- We document the *profit-lock paradox*: in binary markets, profit-taking strategies systematically destroy value because they truncate asymmetric upside while preserving full downside.
- We provide evidence for a two-tier efficiency structure in Polymarket weather markets, with the caveat that elite wallet performance may reflect latency arbitrage rather than superior forecasting.

2 Related Work

2.1 Prediction Markets and Efficiency

Prediction markets are often evaluated as information aggregators, but efficiency varies substantially by contract class [Le, 2026]. Work on prediction markets emphasizes calibration, incentive design, and the relationship between prices and beliefs [Gelman et al., 2020, Lockhart, 2025]. Weather contracts differ from political or sports markets because public forecasts are revised repeatedly on known schedules. Contract semantics—thresholds, stations, and resolution sources—can alter outcome mapping [Gebele and Matthes, 2026].

2.2 Algorithmic Trading and Deployment Gaps

Backtests systematically flatter strategies that fail under realistic execution. Studies identify slippage, latency, fill uncertainty, and nonstationarity as major sources of optimism [Olorunnimbe and Viktor, 2022, Sokolovsky and Arnaboldi, 2023, Pham et al., 2026, Yu et al., 2025]. Liu [2026] document a strikingly similar collapse in Polymarket’s BTC 5-minute contracts: paper trading

yielded $522\times$ returns while live trading produced -49.5% . Dynamic benchmark environments stress temporal ordering and leakage prevention [Liu et al., 2022, 2024].

2.3 Ensemble Forecasting and Trader Heterogeneity

Ensemble methods are natural for weather-linked markets [Mienye and Sun, 2022]. Hybrid forecasting pipelines mixing crowd judgment with machine-generated predictions have become common [Halawi et al., 2024, Chandak et al., 2025, Jin et al., 2026, Schoenegger et al., 2025]. Financial machine learning has examined behavioral and market-state features [Bartram et al., 2020, Mirete-Ferrer et al., 2022, Elebe et al., 2022], but these literatures do not jointly model contract resolution semantics and wallet identity in prediction markets.

3 Data and Methods

3.1 Polymarket Weather Contracts

Polymarket weather contracts are binary options on whether a city’s high temperature will fall within a specific 2°F (US) or 1°C (international) bucket on a given date. Each event offers 8–15 mutually exclusive buckets summing to $\sim\$1.00$. Contracts resolve based on Weather Underground’s reported temperature. Market liquidity ranges from $\$500$ – $\$5,000$ per bucket.

3.2 Dataset

Our analysis spans 30 days of development and 6 weeks of live deployment (February–March 2026):

Table 1: Complete dataset overview.

Source	Type	Trades	Period	P&L
v4: Ensemble forecast	Live	38	Feb 19–27	−\$102
v6: Copy-trading (div. NO)	Live	206	Mar 21–31	−\$134
v7: Elite copy-trading	Live	35	Mar 25–27	−\$118
v8: Convergence	Live	230	Mar 27–30	−\$48
Live total		509		−\$401
Paper v7a (hold-to-res)	Paper	127	Mar 18–28	+\$30
Paper v7b	Paper	98	Mar 18–28	+\$24
Paper v7c	Paper	86	Mar 18–28	−\$24
Paper v7d (elite, H2R)	Paper	64	Mar 18–28	+\$96
Paper v7e	Paper	107	Mar 18–28	+\$46
Paper total		482		+\$172
Wallet snapshots	On-chain	37,562 wallets	Mar 11–20	—

3.3 Strategy Evolution

Each strategy generation addressed a specific failure mode of its predecessor:

Table 2: Strategy generations: each addressed the failure mode of the previous.

Ver.	Strategy	Sizing	Exit	Why previous failed
v4	82-model ensemble, Student- t	17% Kelly, \$3–10	Stop-loss, trailing	(<i>Baseline</i>)
v6	Copy smart wallets, diversified NO	\$30/event budget	Hold-to-resolution	Forecast error $\approx$ bucket width
v7	Copy elite wallets only (YES)	\$10–15	Hold-to-resolution	Too many low-quality signals
v8	3+ independent wallet convergence	\$10–15	Hold-to-resolution	Single-wallet noise

4 Results

4.1 Four Strategies, One Outcome

Table 3 summarizes performance across all four live strategy generations.

Table 3: Live trading results by strategy generation. All strategies lost money.

Strategy	Trades	W	L	WR	P&L
v4: Ensemble forecast	38	8	28	22.2%	−\$102
v6: Copy-trading	206	125	73	63.1%	−\$134
v7: Elite copy-trading	35	2	6	25.0%	−\$118
v8: Convergence	230	19	34	35.8%	−\$48
Total	509	154	141	52.2%	−\$401

The progression tells a story of diminishing losses but persistent failure. The v6 copy-trading strategy achieved the highest win rate (63.1%) but still lost \$134 because winning NO positions in diversified baskets paid small amounts while individual bucket losses were larger. The v8 convergence strategy, the most sophisticated approach, reduced losses to \$48 across 230 trades but still could not achieve positive expected value.

Note that the aggregate win rate of 52.2% is misleading—it includes 177 flat trades in v8 (positions entered but closed at break-even) and 27 flat trades in v7. Among trades with non-zero P&L, the effective win rate is 154 wins vs. 141 losses (52.2%), but average win size (\$3.21) is smaller than average loss size (\$5.23), producing negative expected value per trade.

4.2 The Information-Theoretic Limit

For the 14 v4 trades with forecast data, the forecast error distribution reveals a fundamental bound:

Table 4: Forecast error statistics ($n = 14$). The critical finding: $\text{MAE} \approx$ bucket width.

Metric	Value
Mean Absolute Error (MAE)	2.03°F / 1.3°C
Root Mean Square Error	2.50°F
Mean Signed Error (bias)	−0.86°F
Maximum absolute error	5.8°F
Standard bucket width	2°F / 1°C

This is not merely a practical limitation—it represents an information-theoretic

bound. When forecast MAE equals bucket width, no probabilistic edge can emerge regardless of ensemble size. The 82 models share the same initial conditions, the same underlying physics, and the same systematic biases. Adding ensemble members does not help because the error is irreducible through averaging. This insight motivated the pivot from forecast-based (v4) to signal-based strategies (v6–v8), but as our results show, the signal-based approaches encountered their own structural barriers.

Figure 1 visualizes this relationship: the scatter of forecast errors around the diagonal exceeds the bucket width, making reliable bucket-level prediction impossible.

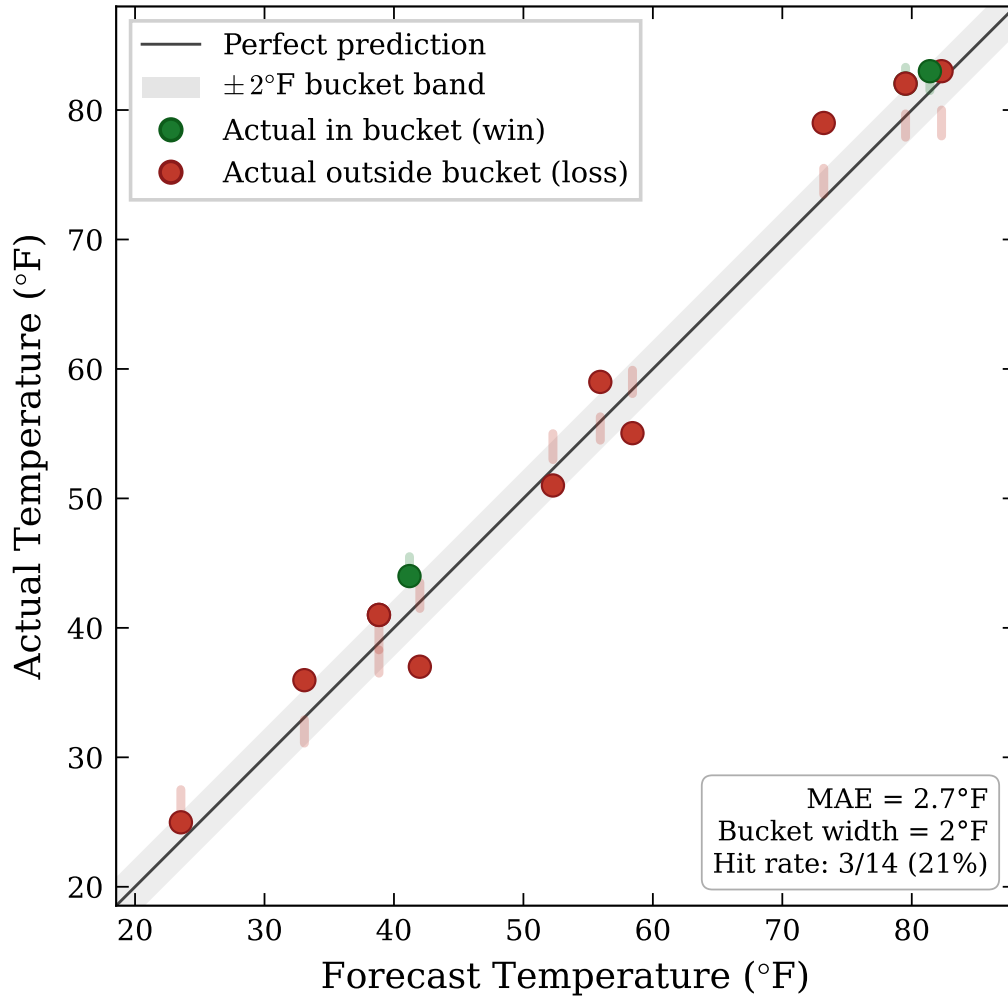


Figure 1: Forecast vs. actual temperature for 14 trades, normalized to Fahrenheit. The gray band shows $\pm 2^\circ\text{F}$ around the perfect prediction line (bucket width). Points outside the band represent forecasts that missed the target bucket. The scatter width approximately equals the bucket width, visualizing the information-theoretic limit.

4.3 The Paper-Trading Mirage

While live trading lost \$401, five paper-trading configurations running on the same markets simultaneously earned \$172:

Table 5: Paper vs. live performance. Paper v7d (elite, hold-to-resolution) was the most profitable paper strategy but could not be replicated live.

Configuration	Trades	WR	P&L	Hold-to-Res	Elite only
Paper v7d	64	79.7%	+\$96	✓	✓
Paper v7e	107	73.8%	+\$46	✓	
Paper v7a	127	78.0%	+\$30	✓	
Paper v7b	98	69.4%	+\$24	✓	
Paper v7c	86	52.3%	−\$24		
Paper total	482	71.0%	+\$172		

The gap between paper and live performance is stark: 71.0% vs. 52.2% win rate, +\$172 vs. −\$401 P&L. This \$573 divergence on the same underlying markets confirms that backtest-to-live degradation in prediction markets is structural, not incidental [Sokolovsky and Arnaboldi, 2023, Pham et al., 2026, Liu, 2026].

4.4 The Profit-Lock Paradox

The paper-trading results reveal a counterintuitive pattern that generalizes to all binary markets. Among the five configurations, the four profitable ones all use hold-to-resolution exits. The one losing configuration (v7c, −\$24) uses intermediate profit-taking.

The mathematical explanation is simple but consequential. Consider a binary position purchased at \$0.20:

- **Hold to resolution:** pays \$0.80 profit (if correct) or −\$0.20 loss. Reward/risk ratio: 4:1.
- **50% profit-lock (exit at \$0.30):** captures \$0.10 gain or −\$0.20 loss. Reward/risk ratio: 1:2.
- **At the limit:** profit-locking captures 12.5% of maximum gain while retaining 100% of maximum loss.

This asymmetry is unique to binary markets. In continuous-return markets, profit-taking is rational because returns are mean-reverting. In binary markets, returns are discontinuous (\$0 or \$1), and early exit systematically converts large potential gains into small realized gains while preserving full downside. **Any prediction market strategy using stop-losses or profit-locks is structurally disadvantaged relative to hold-to-resolution.**

The broader backtest analysis (34 configurations over 178 historical trades) confirms this: only 4 of 34 configurations produced positive returns, all using hold-to-resolution. The strategy with the highest win rate (70.2%, profit-lock at 30%) produced the second-largest loss (−\$92).

4.5 Market Crowding and Edge Erosion

Our wallet tracking system classified 37,562 unique wallets over 10 days. Figure 2 shows the ecosystem dynamics:

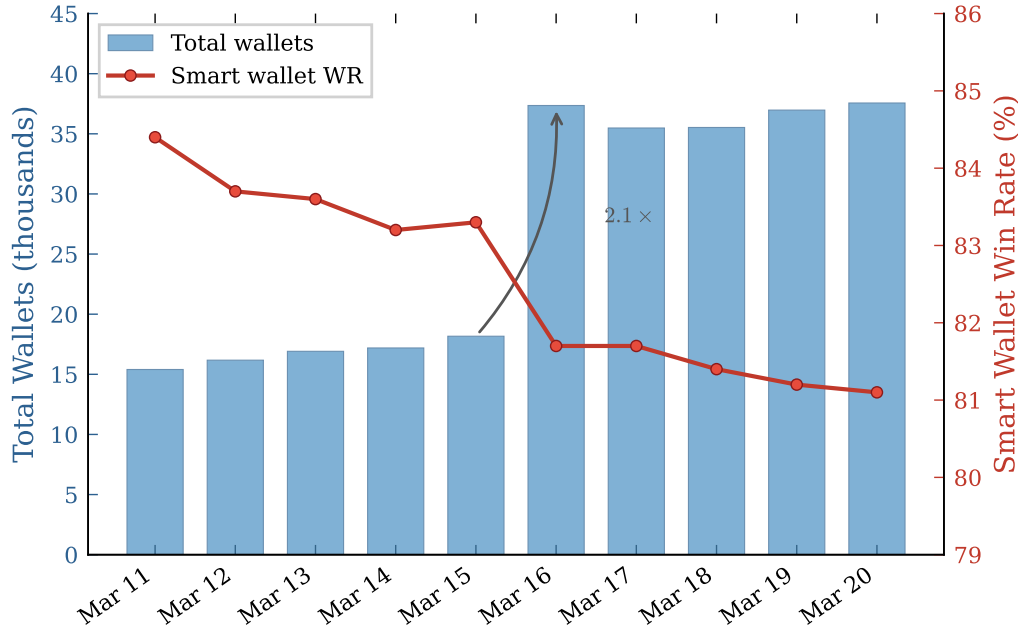


Figure 2: Market crowding dynamics (March 11–20, 2026). Total wallets (bars, left axis) doubled from 15,400 to 37,562 while smart wallet win rates (line, right axis) declined from 84.4% to 81.1%. The sharp jump on March 16 suggests a wave of automated trading bots entering the market.

The data reveals three phenomena:

1. Elite wallet dominance. The top 276 wallets (0.73%) maintain 96.4–97.8% win rates. The top wallet achieved 100% on 26 trades with \$2,185 average size ($p < 10^{-7}$ under chance).

2. Rapid crowding. Wallets grew 144% in 10 days. As of early 2026, 14 of the 20 most profitable Polymarket wallets are automated bots, and arbitrage windows have compressed from 12.3 seconds (2024) to 2.7 seconds.

3. Edge erosion. Smart wallet win rates declined from 84.4% to 81.1% while elite rates remained stable, consistent with crowding that erodes mid-tier edge while leaving structural advantages intact.

Alternative interpretation: latency arbitrage. The 97.8% elite win rate may not reflect superior forecasting. If elite wallets trade primarily in the final hours before resolution—when Weather Underground station data already reflects actual temperature—they exploit the delay between observable information and market price incorporation. Known profitable bots such as “meropi” directly monitor the resolution source in real time. This would still be profitable and inaccessible to our system, but represents latency arbitrage on publicly observable data rather than genuine predictive superiority. Distinguishing these explanations requires block-level timing analysis of elite wallet transactions, which our current dataset does not support.

4.6 Resolution Source Mismatch

A second structural barrier compounds the forecast-precision problem. Our system calibrates against NWS official forecasts, but Polymarket resolves against Weather Underground station observations. These sources diverge by 1–3°F for the same city and date. At 2°F bucket widths, this mismatch can flip the entire win probability for a given contract. A software defect (“phantom edge bug”) additionally inflated NO-side edge signals during the v4 deployment; fixing this

improved backtest P&L by \$29 but represents a correctable engineering error rather than a structural market limitation.

5 Discussion

5.1 Two-Tier Efficiency

Our results support a two-tier efficiency model aligned with domain-specific calibration dynamics [Le, 2026]:

Tier 1 (Retail/Systematic): For participants using publicly available forecasts and on-chain analysis, the market is effectively efficient. Four distinct strategies with 509 combined trades produced negative returns despite progressive sophistication.

Tier 2 (Elite): A small cohort (0.73% of wallets) achieves near-perfect win rates through structural advantages. The evidence for Tier 2 is inferential—we observe outcomes, not strategies—and the latency-arbitrage interpretation weakens the “superior forecasting” framing while preserving the two-tier structure [Bartram et al., 2020].

5.2 Implications for Prediction Market Trading

1. **Never use profit-locks in binary markets.** Hold to resolution or do not enter. This applies to all prediction markets, not only weather contracts.
2. **Calibrate against the resolution source,** not against forecast providers.
3. **Discount backtests by 40–60%** for prediction markets—far more than the 15–30% typical for equities.
4. **Edge erodes in weeks.** The 144% growth in 10 days suggests any publicly identifiable edge has a half-life measured in days, not months.

6 Limitations

- **Strategy-specific sample sizes vary.** While the aggregate of 509 live trades is statistically meaningful, individual strategy generations range from 35 (v7) to 230 (v8) trades. The v7 results in particular should be interpreted with caution.
- **No block-level timing analysis.** We cannot definitively distinguish latency arbitrage from superior forecasting among elite wallets. On-chain transaction timestamps would resolve this.
- **Single starting balance.** All strategies operated on a \$372 account, constraining position sizes to \$3–30. Larger accounts might achieve better fill rates and reduced execution friction.
- **Sequential deployment.** Strategies ran sequentially, not in parallel. Market conditions differed across deployment windows, confounding direct strategy comparison.
- **Wallet observation window.** The 10-day wallet dataset captures a snapshot; longer-term stability of the two-tier structure is unverified.

7 Conclusion

We deployed four algorithmic trading strategies on Polymarket weather markets over six weeks. All four lost money: ensemble forecasting (−\$102), copy-trading (−\$134), elite replication

(−\$118), and convergence signals (−\$48). The total loss of \$401 across 509 trades stands against \$172 in profits from 482 paper trades running on the same markets.

The failure has three structural causes. First, forecast error equals bucket width, creating an information-theoretic bound that no ensemble approach can overcome. Second, any exit strategy short of hold-to-resolution destroys value in binary markets—a paradox that is mathematically inevitable but widely ignored in practice. Third, rapid market professionalization compresses edge windows faster than strategies can adapt.

Yet exploitable edge clearly exists. The 276 elite wallets with 97.8% win rates demonstrate that the market is not uniformly efficient—it is efficient for retail participants using public information, while remaining exploitable through structural advantages in data access, execution speed, and capital scale.

The forensic lesson is not that prediction markets are unbeatable in theory, but that they are unbeatable *in practice* for participants without proprietary infrastructure. For the growing community of algorithmic prediction market traders, this autopsy provides a concrete answer: four strategies, 562 trades, zero edge.

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